Utsav Chatterjee

Software Engineering Intern | AI & Data Science

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https://drylegend.github.io/portfolio/

Professional Summary

Artificial Intelligence and Data Science student at REVA University with experience building machine learning pipelines, NLP applications, and data-driven analytics systems. Skilled in Python, ML model development, and automation workflows with exposure to big data processing and full-stack deployment. Experienced in developing practical AI solutions that solve real-world problems using data analytics, automation, and intelligent systems. Strong analytical mindset with experience collaborating on technical projects, research-oriented analytics, and AI-based applications.

Key Skills / Technical Proficiencies

- **Technical Skills**

Programming: Python, SQL, Java (OOP), JavaScript (basic), C

AI / Machine Learning: Machine Learning, Deep Learning Fundamentals, Natural Language Processing (NLP), Scikit-learn, Predictive Analytics

Data Engineering & Analytics: Pandas, NumPy, Apache Spark, Hadoop, Data Visualization, Power BI

Development & Tools: Git, GitHub Actions (CI/CD), Docker (basic), Jupyter Notebook, FastAPI

Platforms: Linux, Windows

- **Soft Skills**

Collaboration, Problem-Solving, User-Centered Design, Process Optimization, Documentation, Project Management, Adaptability

Work Experience

- **Teaching Assistant Intern — RACE (REVA Academy for Corporate Excellence)**

  Bangalore, India | 2024 – Present

  - Assisted faculty in delivering AI/ML and Data Science lab sessions for undergraduate students.
  - Guided students in implementing Python-based machine learning algorithms and debugging code.
  - Supported hands-on projects involving data preprocessing, ML models, and analytics workflows.
- ***Production & Testing Coordinator — Centum Electronics Ltd Bangalore, India | 2017–2022***
  - Reduced electronic testing time from 24 hours to 8 hours through process optimization and workflow redesign.
  - Increased production throughput from 300 to 2000 boards per month, improving operational efficiency.
  - Implemented structured defect documentation and coordinated technicians to ensure quality and delivery targets.

Projects

1. **NOVAGENT: A Multi-Agent Query-Aware Research Retrieval System with Citation Graph-Based Novelty Modelling (Ongoing Research Project)**
   - Designing a pipeline to extract and analyze research paper metadata and content.
   - Implementing semantic similarity and topic mapping to link related research works.
   - Building a prototype interface to explore research connections interactively.

   Tech: Python, NLP, embeddings, data processing, visualization
2. **Automated Weather ML Pipeline (MLOps Project)**
   - Designed an automated ML pipeline with CI/CD integration using GitHub Actions.
   - Implemented data preprocessing, automated model training, and retraining workflows for updated datasets.
   - Enabled scalable ML experimentation and automated deployment of models.

   Tech: Python, GitHub Actions, CI/CD, ML
3. **Global Temperature Analysis using Apache Spark (BDA Lab)**
   - Built a Spark-based analytics pipeline to analyze 160+ years of global climate data.
   - Performed trend detection, anomaly identification, and geospatial analysis.
   - Optimized Spark transformations achieving ~40% faster processing.

   Tech: PySpark, Spark SQL, Hadoop, Python, Power BI

4. **Big Data Crime Analytics Dashboard**
   - Developed a multi-page Power BI dashboard to analyze cybercrime datasets and identify trends.
   - Implemented KPIs, interactive filters, and data visualization for decision-support insights..

   Tech: Power BI, Data Cleaning, Visualization

Certifications & Training

- Data Structures & Algorithms (C) – Academic
- Big Data Analytics – Academic
- IBM NLP Certificate – IBM
- Cloud Computing Fundamentals – IBM

Education

1. ***BTech in Arti□icial Intelligence & Data Science***
   REVA UNIVERSITY | Bangalore, India | 2023 – Present
   Relevant Coursework: Data Structures (C), Cloud Computing, Big Data Analytics, NLP, Deep Learning, AI/ML Foundations, DAA, Pattern Recognition
2. ***Diploma in Electrical Engineering***
   South Calcutta Polytechnic | 2014 – 2017
3. ***Higher Secondary in Panskura B.B. High School in Science 2012-2014***

Awards / Publications

- Reduced testing time from **24 hrs → 8 hrs**, increasing throughput from **300 → 2000** units/month at Centum Electronics got Appreciation Award.
- Represented in multiple hackathons including CodeSprint 3.0, Ideathon(Figma Design)
- Built multiple academic and portfolio-grade ML/AI systems
- MS Excel from Simplilearn